

ENTHUSIAST ADVANCE COURSE

PHASE : MEC-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 04-03-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	3	3	3	1	4	3	3	2	2	2	1	2	2	3	2	1	4	2	1	2	3	3	2	1	3	3	1	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	3	2	2	4	1	1	3	1	3	2	1	3	2	3	2	4	3	1	3	4	2	4	1	2	2	4	3	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	4	3	3	2	2	2	2	3	2	2	1	4	1	1	2	2	3	2	3	3	3	3	3	2	2	3	1	4	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	4	2	1	4	4	2	4	2	3	3	4	2	4	2	3	4	2	4	2	2	1	1	1	1	1	3	3	1	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	3	4	3	3	2	2	3	1	1	4	4	4	4	3	3	1	2	1	1	4	1	4	3	4	2	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	3	3	1	1	2	3	1	3	1	4	3	2	4	2	1	2	1	1	2	4	4	3	4	3	3	3	4	1	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	1	3	2	2	3	4	3	3	3	4	4	4	2	1	1	4	2	1	4										

HINT - SHEET

SUBJECT : PHYSICS

SECTION - A

1. **Ans (2)**

$$W_1 = U_f - U_i = -\frac{GMm}{R} - \left(-\frac{3}{2}\frac{GMm}{R}\right) = \frac{GMm}{2R}$$

$$W_2 = U_f - U_i = 0 - \left(-\frac{GMm}{R}\right) = \frac{GMm}{R}$$

$$\frac{W_1}{W_2} = \frac{\frac{GMm}{2R}}{\frac{GMm}{R}} = \frac{1}{2}$$

2. **Ans (2)**

$$-\frac{GMm}{R} + \frac{1}{2}m\left(\frac{3}{4}V_e\right)^2 = -\frac{GMm}{R+h} + 0$$

$$-\frac{GM}{R} + \frac{1}{2} \times \frac{9}{16} \times \frac{2GM}{R} = -\frac{GM}{R+h}$$

$$\frac{-7GM}{16R} = -\frac{GM}{R+h}$$

$$7R + 7h = 16R$$

$$h = \frac{9R}{7}$$

3. **Ans (3)**

$$W_d = \frac{1}{4} \times w$$

$$mg_d = \frac{1}{4} \times mg$$

$$g\left(1 - \frac{d}{R}\right) = \frac{g}{4}$$

$$1 - \frac{d}{R} = \frac{1}{4}$$

$$\frac{d}{R} = \frac{3}{4} \Rightarrow d = \frac{3}{4}R$$

4. **Ans (3)**

$$E = \frac{GMm}{2r}$$

$$L = mv_0r = m\sqrt{\frac{GM}{r}}r$$

$$L = \sqrt{\frac{GMm^2r^2}{r}}$$

$$L = \sqrt{\frac{2GMm}{2r}} \times mr^2$$

$$L = \sqrt{2Emr^2}$$

7. **Ans (4)**

$\Delta P = \text{area under the F-t graph}$

$$\Delta P = \frac{1}{2} \times 8 \times 1 - 2 \times 0.5 = \frac{3\text{kg} - \text{m}}{\text{s}}$$

10. **Ans (2)**

$$\Delta U = U_f - U_i$$

$$= \left(\frac{-GMm}{4R} \right) - \left(\frac{-GMm}{R} \right)$$

$$= \frac{-GMm}{4R} + \frac{GMm}{R}$$

$$= \frac{3GMm}{4R}$$

$$\Delta U = \frac{3}{4}mgR \quad \left(g = \frac{GM}{R^2} \right)$$

11. **Ans (2)**

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$\left(\frac{T_A}{T_B} \right)^2 = \left(\frac{r_1}{r_2} \right)^3$$

$$\left(\frac{T_A}{T_B} \right)^2 = \left(\frac{4}{1} \right)^3$$

$$\frac{T_A}{T_B} = 8$$

12. **Ans (2)**

Acceleration due to gravity at a height h from earth's surface

$$g_h = \frac{g}{\left(1 + \frac{h}{R} \right)^2}$$

$$\frac{g}{100} = \frac{g}{\left(1 + \frac{h}{R} \right)^2}$$

$$1 + \frac{h}{R} = 10$$

$$h = 9R$$

15. **Ans (2)**

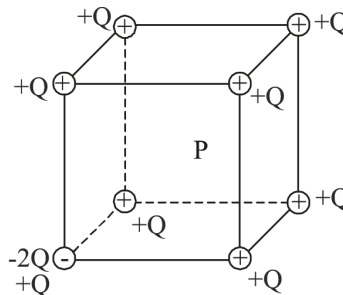
TG: @Chalnaayaaar

$$E_1 = \frac{kQ}{r^2}; E_2 = \frac{2k\lambda}{r} = \frac{2k}{r} \left(\frac{Q}{\pi r} \right)$$

$$E_2 = \frac{2kQ}{\pi r^2}$$

$$\text{So } E_1 > E_2$$

16. **Ans (3)**



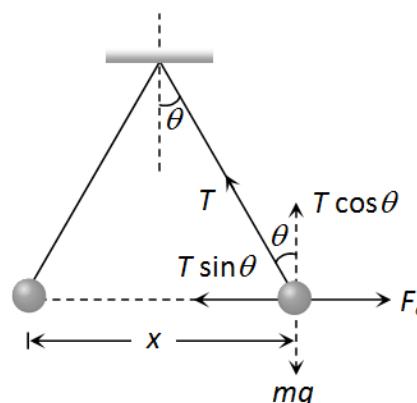
By symmetry, electric field at P due to all shown positive charges is zero.

$$\text{So } E_p = \frac{2kQ}{r^2}$$

17. **Ans (2)**

Opposite charge of rod is induced on water and it gets attracted.

18. **Ans (1)**



$$\text{In equilibrium } F_e = T \sin \theta \quad \dots\dots (i)$$

$$mg = T \cos \theta \quad \dots\dots (ii)$$

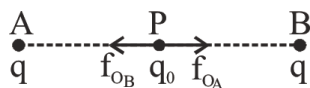
$$\tan \theta = \frac{F_e}{mg} = \frac{q^2}{4\pi\epsilon_0 x^2 \times mg} \text{ also } \tan \theta \approx \sin \theta = \frac{x/2}{L}$$

$$\text{Hence } \frac{x}{2L} = \frac{q^2}{4\pi\epsilon_0 x^2 \times mg}$$

$$\Rightarrow x^3 = \frac{2q^2 L}{4\pi\epsilon_0 mg} \Rightarrow x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$$

19. Ans (4)

case I :



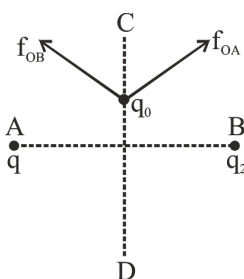
$$f_{OB} = f_{OA}$$

when q_0 is displaced slightly towards charge

B. $f_{OB} > f_{OA}$

Hence stable equilibrium

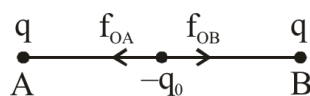
case II :



$\vec{f}_{OA} + \vec{f}_{OB}$ is along DC towards C

$\Rightarrow$ Unstable equilibrium

case III :



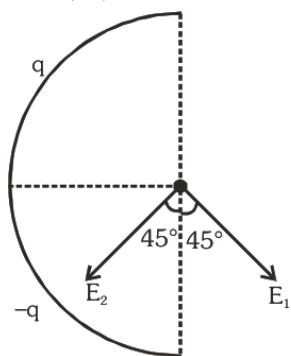
$$f_{OA} = f_{OB}$$

If $-q_0$ is displaced slightly towards B.

$\Rightarrow f_{OB} > f_{OA}$

$\Rightarrow$ Unstable equilibrium

20. Ans (2)



$$E_1 = E_2 = \frac{2k\lambda \sin\left(\frac{90}{2}\right)}{r} = \frac{2k\lambda}{\sqrt{2}r}$$

and $E_{net} = 2E_1 \cos 45^\circ$

$$= 2 \left(\frac{2k\lambda}{\sqrt{2}r} \right) \times \frac{1}{\sqrt{2}} = \frac{4kq}{\pi r^2}$$

$$\text{Here } k = \frac{1}{4\pi\epsilon_0}$$

21. Ans (1)

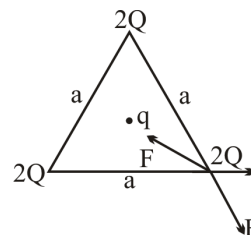
TG: @Chalnaayaaar

$$q_{\text{total}} = \int_0^R \sigma ds = \int_0^R (\sigma_0 r^3) (2\pi r dr)$$

$$= 2\pi\sigma_0 \int_0^R r^4 dr = 2\pi\sigma_0 \left[\frac{r^5}{5} \right]_0^R$$

25. Ans (2)

q should be negative



$$F' = \sqrt{3}F$$

$$\frac{2KQq}{(a/\sqrt{3})^2} = \sqrt{3} \frac{4KQ^2}{a^2} \quad (\because q \text{ should be negative})$$

$$q = -\frac{2Q}{\sqrt{3}}$$

26. Ans (1)

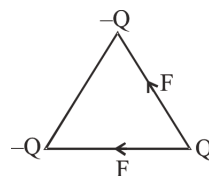
$$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$$

Max. electric field at $x = \frac{R}{\sqrt{2}}$

$$E = \frac{kQ \frac{R}{\sqrt{2}}}{\left(R^2 + \frac{R^2}{2}\right)^{3/2}} = \frac{kQR}{\sqrt{2} \left(\frac{3R^2}{2}\right)^{3/2}}$$

$$= \frac{2kQR}{3\sqrt{3}R^3} = \frac{2kQ}{3\sqrt{3}R^2}$$

27. Ans (3)



$$F_{\text{net}} = \sqrt{3}F = \sqrt{3} \frac{kQ^2}{a^2}$$

28. Ans (3)

$$mr = \left(\frac{2\pi}{t} \right)^2 = \frac{Ke^2}{r^2}$$

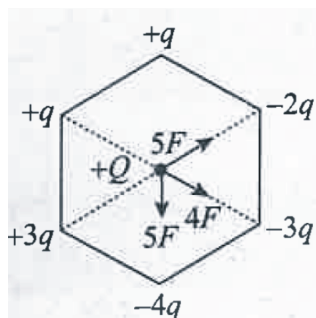
$$T_2 \propto r^3$$

$$T \propto r^{3/2}$$

29. Ans (1)

$$E = \frac{K(3q)}{a^2}(-\hat{j}) + \frac{Kq}{a^2}(\hat{j}) = \frac{-2Kq}{a^2}(\hat{j})$$

30. Ans (4)



$$F_{\text{net}} = 4F + \sqrt{25F^2 + 25F^2 + 50F^2 \cos 120^\circ}$$

$$\Rightarrow F_{\text{net}} = 9F$$

31. Ans (4)

$$F_1 = \frac{kq_1 q_2}{(\ell \sin 30^\circ)^2}$$

$$F_2 = \frac{kq_1 q_3}{(\ell \sin 30^\circ)^2}$$

$$\text{So, } \frac{F_1}{F_2} = \frac{q_2 (\cos 30^\circ)^2}{q_3 (\sin 30^\circ)^2} = \frac{3}{2} \times (\sqrt{3})^2 = \frac{9}{2}$$

35. Ans (2)

$$U = \frac{A\sqrt{x}}{x^2 + B}$$

$$B = L^2$$

$$ML^2 T^{-2} \times L^2$$

$$\frac{ML^4 T^{-2}}{L^{1/2}} = A$$

$$ML^{4-\frac{1}{2}} T^{-2} = A$$

$$[ML^{7/2} T^{-2}] = [A]$$

SECTION - B

36. Ans (4)

The time period of moon's revolution around the earth does not depend upon the mass of the moon.

37. Ans (1)

$$g_m = \frac{1}{6} g_e$$

$$\frac{4}{3} \pi R_m \rho_m G = \frac{1}{6} \times \frac{4}{3} \pi R_e \rho_e G$$

$$R_m = \frac{\rho_e}{6\rho_m} R_e = \frac{5}{3} \times \frac{1}{6} \times R_e$$

$$R_m = \frac{5}{18} R_e$$

39. Ans (3)

$$t = \frac{V - U}{a}, \left\{ a = -\frac{5}{5} = -1 \right\}$$

$$t = \frac{0 - 40}{-1} = 40 \text{ sec.}$$

40. Ans (1)

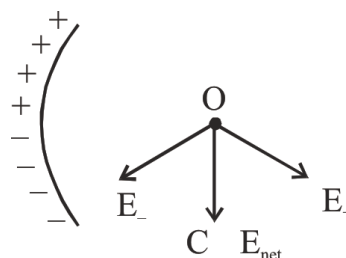
$$-\frac{3}{2} \frac{GMm}{R} + \frac{1}{2} mv^2 = -\frac{GMm}{R}$$

$$\Rightarrow v = \sqrt{\frac{GM}{R}}$$

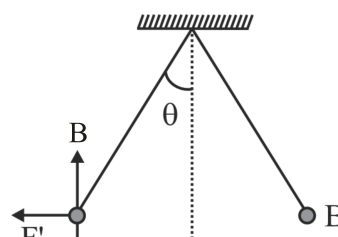
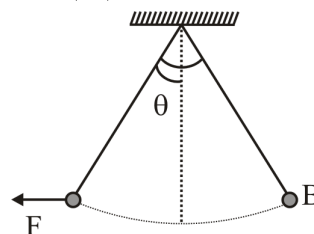
41. Ans (3)

Given diagram shows :

The direction of E_{net} is along OC.



42. Ans (2)



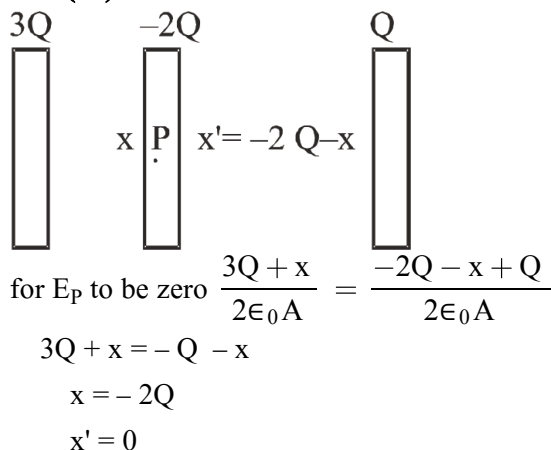
$$\tan \theta = \frac{F_e}{F_{\text{gravity}}}$$

If θ does not change then

$$\frac{q^2}{\frac{4\pi\epsilon_0 r^2}{mg}} = \frac{q^2}{\frac{4\pi\epsilon_0 k r^2}{mg - B}}$$

$$K = \frac{mg}{mg - B} = \frac{V \rho g}{V \rho g - v d g} = \frac{16}{16 - 8} = 2$$

43. Ans (1)



44. Ans (3)

Use

$$\frac{kq}{r^2}$$

46. Ans (3)

Enter Flux

$$\phi_{\text{Enter}} = E_1 A \cos 180^\circ$$

$$= (\alpha\sqrt{a})(a^2)(-1)$$

$$\phi_1 = -\alpha a^{5/2}$$

Exit Flux

$$\phi_{\text{exit}} = E_2 A \cos 0^\circ$$

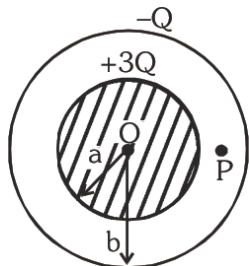
$$= (\alpha\sqrt{2a})(a^2) \times 1$$

$$\phi_2 = \sqrt{2}\alpha a^{5/2}$$

$$\text{Net Flux} = \phi_2 - \phi_1$$

$$\phi_{\text{Net}} = \phi_{\text{Net}} = (\sqrt{2} - 1) \alpha a^{5/2}$$

47. Ans (2)

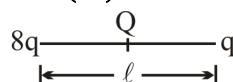


Electric field at point 'P' due to spherical shell is zero and due to solid sphere.

$$E_P = \frac{k3Q}{R^2} = \frac{3Q}{4\pi\epsilon_0 R^2}$$

48. Ans (4)

TG: @Chalnaayaaar



4q will repel q, so Q must be negative to attract q

$$F_{q,Q} \leftarrow q \rightarrow F_{q,4q} \quad F_{q,Q} = F_{q,4q}$$

$$\Rightarrow \frac{KQq}{(\ell/2)^2} = \frac{K(8q)(q)}{\ell^2}$$

$$Q = -2q$$

SUBJECT : CHEMISTRY

SECTION-A

60. Ans (4)

For reactant 'P'

$$\text{Since } t_{75\%} = 2 t_{50\%}$$

therefore order with respect to 'P' is 'one'

For reactant 'Q'

linear variation of concentration of 'Q' with time suggest that order with respect to 'Q' is

zero therefore overall order will be $= 1 + 0 = 1$

Correct Option (D)

(Assuming conc. of other reactant to be very large)

63. Ans (3)

$$\Delta H = (E_a)_f - (E_a)_b$$

$$5 = 15 - (E_a)_b$$

$$(E_a)_b = 15 - 5 = 10 \text{ kJ mol}^{-1}$$

SUBJECT : BIOLOGY-I

SECTION-A

101. Ans (3)

NCERT Pg. # 147

104. Ans (4)

NCERT XII, Pg # 159

105. Ans (2)

NCERT XII, Pg # 154

106. Ans (3)

NCERT XII, Pg # 152

110. Ans (2)

NCERT (H) pg.# 163

114. Ans (1)
NCERT (XII) Page # 147 (E)

116. Ans (1)
NCERT XII, Pg. # 153 (E), 167 (H)

SECTION-B

141. Ans (1)
NCERT Pg#147

142. Ans (1)
NCERT Pg#158

SUBJECT : BIOLOGY-II

SECTION-A

151. Ans (4)
NCERT XII Pg # 251, 252

153. Ans (3)
NCERT XII, Page # 234

154. Ans (1)
NCERT-XII, Page # 250

157. Ans (3)
Module 3 Pg. # 3

160. Ans (1)
NCERT-XII, Pg. # 235

163. Ans (2)
NCERT Pg. # 237, 238

166. Ans (1)
NCERT Pg. # 228

167. Ans (2)
NCERT Pg. # 232

169. Ans (1)
NCERT Pg#220

170. Ans (2)
NCERT Pg#220

171. Ans (4)
NCERT Pg#221

172. Ans (4)
NCERT Pg#221

173. Ans (3)
NCERT Pg#228

174. Ans (4)
NCERT Pg#229-330

175. Ans (3)
NCERT Pg#227

176. Ans (3)
NCERT Pg#227

177. Ans (3)
NCERT Pg#231

178. Ans (4)
NCERT Pg#233

179. Ans (1)
NCERT Pg#233

180. Ans (2)
NCERT Pg#233

181. Ans (1)
NCERT Pg#235

182. Ans (1)
NCERT Pg#21

183. Ans (3)
NCERT Pg#237

184. Ans (2)
NCERT Pg#237

SECTION-B**188. Ans (3)**

NCERT Pg.# 234

193. Ans (4)

NCERT Pg# 227,229

194. Ans (2)

NCERT Pg#230

195. Ans (1)

NCERT Pg#227

196. Ans (1)

NCERT Pg#229

197. Ans (4)

NCERT Pg#230

198. Ans (2)

NCERT Pg#231

199. Ans (1)

NCERT Pg#235